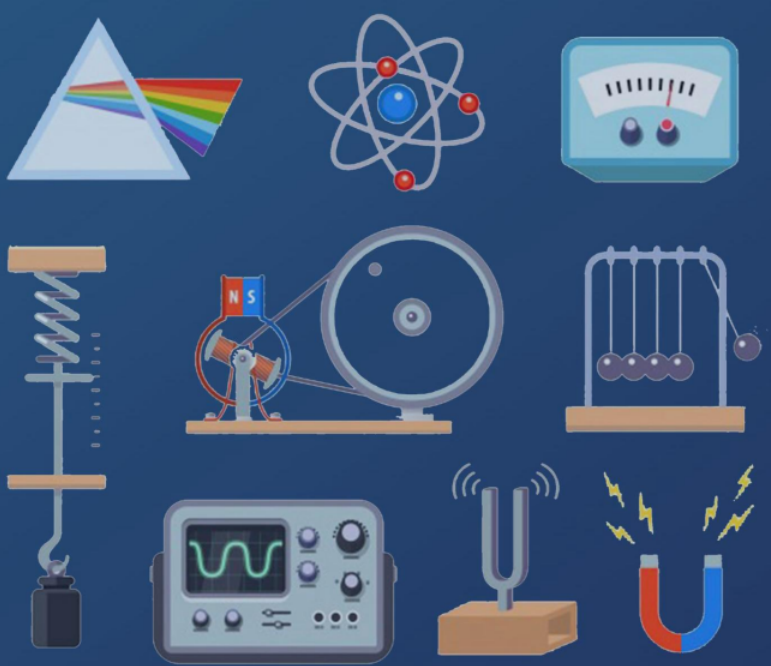
PHYSICS

Thermodynamics



SHEET

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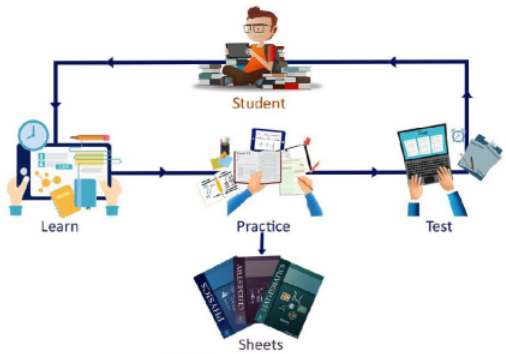
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Work Done By Gas	1	17	1	1
2	First Law of Thermodynamics	2,3	1	2,3	2,3
3	Isochoric and Isobaric Process	4-10	2-4, 18,19	4	4,17
4	Isothermal Process		5,6	5-7	
5	Adiabatic Process	11,12	7,8	8-12	5,18
6	Polytropic Process and Free Expansion of Gas		9,15	13-15	6,7,22
7	Miscellaneous Practice Qs	13-16	16	16-19	8-11, 16,19
8	Carnot Engine And Second Law of Thermodynamics	17	10,12-14		20
9	Refrigerators And Heat Pumps	18	20	20,21	12,21
10	After Full Chapter	19	11	22	13-15

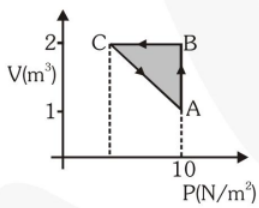
Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Work Done By Gas

Q1. An ideal gas is taken through the cycle $A \to B \to C \to A$, as shown in the figure. If the net heat supplied to the gas in the cycle is 5 J, the work done by the gas in the process $C \to A$ is :–



(A) –5J (B) –10J

(C) –15J (D) –20J

First Law of Thermodynamics

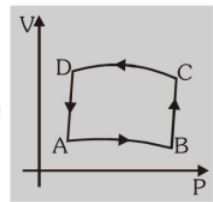
Q2. 1 kg of a gas does 20 kJ of work and receives 16 kJ of heat when it is expanded between two states. A second kind of expansion can be found between the same initial and final state which requires a heat input of 9 kJ. The work done by the gas in the second expansion is :

(A) 32 kJ (B) 5 kJ

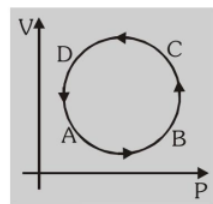
(C) – 4 kJ (D) 13 kJ

Q3. In the diagrams (i) to (iv) of variation of volume with changing pressure is shown. A gas is taken along the path ABCDA. The change in internal energy of the gas will be:-

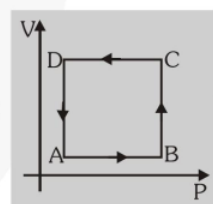
(i)



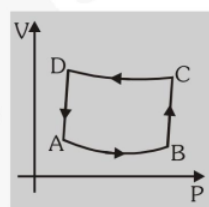
(ii)



(iii)



(iv)



(A) Positive in all cases (i) to (iv)

(B) Positive in cases (i), (ii) and (iii) but zero in case (iv)

(C) Negative in cases (i), (ii) and (iii) but zero in case (iv)

(D) Zero in all the four cases

Isochoric and Isobaric Process

Q4. C_P is always greater than C_V due to the fact that :–

(A) No work is being done on heating the gas at constant volume.

(B) When a gas absorbs heat at constant pressure its volume do not change so as to do some external work.

(C) The internal energy is a function of temperature only for an ideal gas.

(D) For the same rise of temperature, the internal energy of a gas changes by a smaller amount at constant volume than at constant pressure.

Q5. **Statement–1 :** In a process if initial volume is equal to the final volume, work done by the gas is zero.

Statement–2 : In an isochoric process work done by the gas is zero.

(A) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1.

(B) Statement–1 is True, Statement–2 is True; Statement–2 is not a correct explanation for Statement–1.

(C) Statement–1 is True, Statement–2 is False.

(D) Statement–1 is False, Statement–2 is True.

Q6. If one mole of a mono-atomic gas $\left[\gamma = \frac{5}{3}\right]$ is mixed with one mole of a diatomic gas $\left[\gamma = \frac{7}{5}\right]$, the value of γ for the mixture is :-

(A) 1.4 (B) 1.5 (C) 1.53 (D) 3.07

Q7. For a gas if $\gamma = 1.4$, then atomicity, C_p and C_v of the gas are respectively :–

(A) monoatomic, $\frac{5}{2}R, \frac{3}{2}R$

(B) monoatomic, $\frac{7}{2}R, \frac{5}{2}R$

(C) diatomic, $\frac{7}{2}R, \frac{5}{2}R$

(D) triatomic, $\frac{7}{2}R, \frac{5}{2}R$

Q8. Correct relation between molar specific heat at constant pressure and constant volume :- (R universal gas constant)

(A) $C_P - C_V = r$

(B) $C_P - C_V = R$

(C) $C_P + C_V = R$

(D) $m\,C_P dt - mC_V dt = R$

Q9. For an ideal monoatomic gas, the universal gas constant R is n times the molar heat capacity at constant pressure C_p. Here n is -

(A) 0.67 (B) 1.4

(C) 0.4 (D) 1.67

Q10. If the ratio of specific heat of a gas at constant pressure to that at constant volume is γ, the change in internal energy of a mole of gas, when the volume changes from V to 2V at constant pressure P, is :-

(A) $R/(\gamma - 1)$ (B) PV

(C) $PV/(\gamma - 1)$ (D) $\gamma PV/(\gamma - 1)$

Adiabatic Process

Q11. The adiabatic Bulk modulus of a diatomic gas at atmospheric pressure is

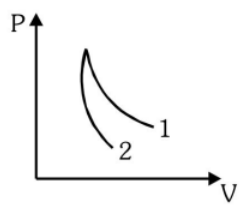
(A) 0 Nm^{-2}

(B) 1 Nm^{-2}

(C) 1.4×10^4 Nm^{-2}

(D) 1.4×10^5 Nm^{-2}

Q12. P–V plots for two gases during adiabatic processes are shown in the figure. Plots 1 and 2 should correspond respectively to :



(A) He and O_2 (B) O_2 and He

(C) He and Ar (D) O_2 and N_2

Miscellaneous Practice Qs

Q13. One mole of an ideal gas at temperature T_1 expands according to the law $\frac{P}{V^2} = a$ (constant). The work done by the gas till temperature of gas becomes T_2 is :

(A) $\frac{1}{2}R(T_2 - T_1)$ (B) $\frac{1}{3}R(T_2 - T_1)$

(C) $\frac{1}{4}R(T_2 - T_1)$ (D) $\frac{1}{5}R(T_2 - T_1)$

Q14. An ideal gas expands isothermally from a volume V_1 to V_2 and then compressed to original volume V_1 adiabatically. Initial pressure is P_1 and final pressure is P_3. The total work done is W. Then -

(A) $P_3 > P_1$, $W > 0$ (B) $P_3 < P_1$, $W < 0$

(C) $P_3 > P_1$, $W < 0$ (D) $P_3 = P_1$, $W = 0$

Q15. Starting with the same initial conditions, an ideal gas expands from volume V_1 to V_2 in three different ways, the work done by the gas W_1 if the process is purely isothermal, W_2 if purely isobaric and W_3 if purely adiabatic, then :-

(A) $W_2 > W_1 > W_3$ (B) $W_2 > W_3 > W_1$

(C) $W_1 > W_2 > W_3$ (D) $W_1 > W_3 > W_2$

Q16. n moles of an ideal gas undergo a process in which the temperature changes with volume as $T = kV^2$. The work done by the gas as the temperature changes from T_0 to $4T_0$ is-

(A) $3nRT_0$ (B) $\left(\frac{5}{2}\right)nRT_0$

(C) $\left(\frac{3}{2}\right)nRT_0$ (D)) Zero

Carnot Engine And Second Law of Thermodynamics

Q17. A Carnot engine, whose efficiency is 40%, takes in heat from a source maintained at a temperature of 500 K. It is desired to have an engine of efficiency 60%. Then, the intake temperature for the same exhaust (sink) temperature must be :

(A) 750 K

(B) 600 K

(C) Efficiency of Carnot engine cannot be made larger than 50%

(D) 1200 K

Refrigerators And Heat Pumps

Q18. An ideal refrigerator has a freezer at a temperature of -13^0C. The coefficient of performance of the engine is 5. The temperature of the air (to which heat is rejected) will be :

(A) 325^0C (B) 325 K

(C) 39^0C (D) 320^0C

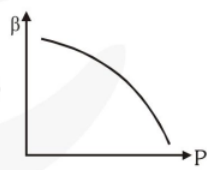
After Full Chapter

Q19. Which of the following graphs correctly represent the variation of $\beta = -\frac{dV/dP}{V}$ with P for an ideal gas at constant temperature ?

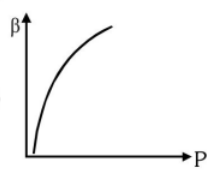
(A)



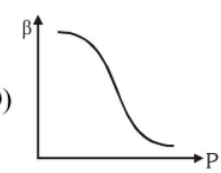
(B)



(C)



(D)



ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. In a given process of an ideal gas, dW = 0 and dQ < 0. Then for the gas :–

(A) the temperature will decrease

(B) the volume will increase

(C) the pressure will remain constant

(D) the temperature will increase

Q2. The specific heats of a gas are C_P=0.2 cal/g °C & C_V = 0.15 cal/g °C.

[Take R=2 cal/mole^{0}C]

(A) The molar mass of the gas is 40 g

(B) The molar mass of the gas cannot be determined from the data given

(C) The number of degrees of freedom of the gas molecules is 6

(D) The number of degrees of freedom of the gas molecules is 8

Q3. Statement A : $C_P - C_V = R$;

Statement B : $\frac{C_P}{C_V} = 1.67$

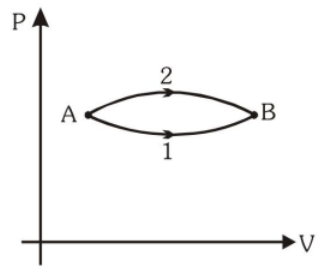
(A) A and B statement are true for all ideal gases.

(B) A is true for all ideal gases and B is true for only monoatomic gases

(C) A and B are true for monoatomic gases.

(D) A is true for all ideal gases and B is true for only diatomic gases

Q4. The figure shows two paths for the change of state of a gas from A to B. The ratio of molar heat capacities in path 1 and path 2 is :-



(A) > 1 (B) < 1

(C) 1 (D) Data insufficient

Q5. In the isothermal expansion of a gas :-

(A) work done on a gas is zero

(B) heat is necessarily absorbed

(C) the energy stored in the gas remains constant

(D) neither heat is supplied to the gas, nor withdrawn

Q6. The process in which the energy given to a system is completely transformed into work is :-

(A) isobaric process

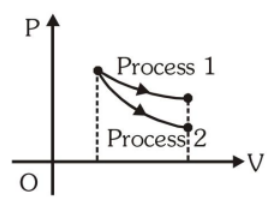
(B) isometric process

(C) isothermal process

(D) adiabatic process

Q7. The indicator diagram for two process 1 and 2 carried on an ideal gas is shown in figure.

If m_1 and m_2 be the slopes $\left(\frac{dP}{dV}\right)$ for process 1 and process 2 respectively, then:–



(A) $m_1 = m_2$ (B) $m_1 > m_2$

(C) $m_1 < m_2$ (D) $m_2 C_V = m_1 C_P$

Q8. An ideal gas is heated from temperature T_1 to T_2 under various conditions. The correct statement(s) is/are :-

(A) $\Delta U = nC_V (T_2 - T_1)$ for isobaric, isochoric and adiabatic process

(B) Work is done at expense of internal energy in an adiabatic process and both have equal values

(C) $\Delta U = 0$ for an isothermal process

(D) $C = 0$ for an adiabatic process

Q9. 2 moles of a monoatomic gas are expanded to double its initial volume, through a process P/V = constant. If its initial temperature is 300 K, then which of the following is not true.

(A) $\Delta T = 900$ K (B) $\Delta Q = 3200R$

(C) $\Delta Q = 3600R$ (D) $W = 900R$

Q10. According to the second law of thermodynamics :

(A) Heat energy cannot be completely converted to work

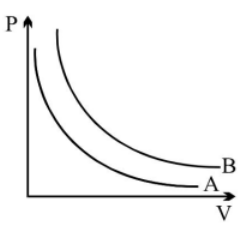
(B) Work cannot be completely converted to heat energy

(C) For all cyclic processes we have $dQ/T < 0$

(D) The reason all heat engine efficiencies are less than 100% is friction, which is unavoidable

(E) All of the above are true

Q11. Figure shows the pressure P versus volume V graphs for two different gas sample at a given temperature. M_A and M_B are masses of two samples, n_A and n_B are numbers of moles. Which of the following ***must be incorrect.***



(A) $M_A > M_B$ (B) $M_A < M_B$

(C) $n_A > n_B$ (D) $n_A < n_B$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 12 & 14

Entropy (S) is a thermodynamic variable like pressure P, volume V and temperature T. Entropy of a thermodynamic system is a measure of disorder of molecular motion. Greater is disorder, greater is entropy. Change in entropy of a thermodynamic system is the ratio of heat supplied to absolute temperature. In an adiabatic reversible process, entropy remains constant while in any irreversible process entropy increases. In nature the processes are irreversible; therefore entropy of universe is continuously increasing.

Q12. The unit of entropy in S–I system is–

(A) cal/K

(B) joule/kg

(C) joule/K

(D) kilocal/°C

Q13. When milk is heated, its entropy :

(A) increases

(B) decreases

(C) remains unchanged

(D) may decrease or increase

Q14. After a long–long time, the energy available for work will be :

(A) as much as present value

(B) much less than present value

(C) much more than present value

(D) can not say

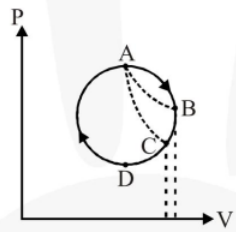
MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q15.

Column–I	Column–II
(A) Isobaric process	(p) No heat exchange
(B) Isothermal process	(q) Constant pressure
(C) Isoentropy process	(r) Constant internal energy
(D) Isochoric process	(s) Work done is zero

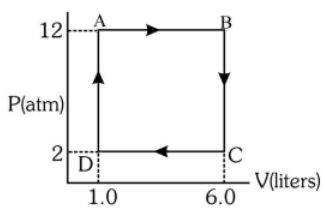
Q16. For an ideal gas a process PV diagram is a circle. An adiabatic from A passes through C. An isotherm from A passes through B. We take a part of the circular cyclic process. Comment on the sign of the quantity of column-I.



Column-I	Column-II
(A) Heat given to the gas in going from A to C along circle	(p) Positive
(B) Heat given to the gas in going from B to C along circle	(q) Negative
(C) Heat given to the gas in going from C to D along circle	(r) Zero
	(s) can't be said

SUBJECTIVE TYPE

Q17. The adjoining diagram shows a P–V graph of the thermodynamic behaviour of an ideal gas. Find out from this graph (i) Work in the processes A → B, B → C, C → D and D → A,



Q18. For a gas $\frac{R}{C_P} = 0.4$. For this gas calculate the following –

(a) atomicity and degree of freedom

(b) value of C_v and γ

(c) mean gram - molecular kinetic energy at 300 K temperature

Q19. At 8 atm pressure a monoatomic gas is expands from 2 litre to 5 litre then calculate the following

(a) Work done by the gas

(b) Increase in internal energy

(c) Amount of heat supplied

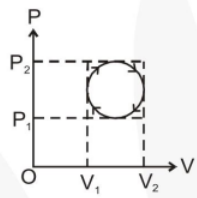
Q20. A refrigerator takes heat from water at 0°C and transfer it to room at 27°C. If 100 kg of water is converted in ice at 0°C then calculate the workdone.

(Latent heat of ice is 3.4×10^5 J/kg)

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. In a cyclic process shown on the P – V diagram the magnitude of the work done is :



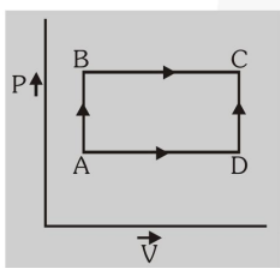
(A) $\pi\left(\frac{P_2 - P_1}{2}\right)^2$

(B) $\pi\left(\frac{V_2 - V_1}{2}\right)^2$

(C) $\frac{\pi}{4}(P_2 - P_1)(V_2 - V_1)$

(D) $\pi(P_2V_2 - P_1V_1)$

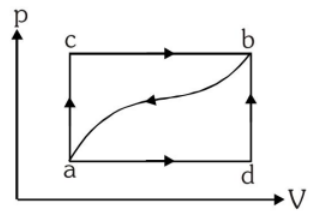
Q2. As shown in the figure, the amount of heat absorbed along the path ABC is 90J and the amount of work done by the system is 30 J. If the amount of work done along the path ADC is 20 J the amount of heat absorbed will be :-



(A) 80 J (B) 90 J

(C) 110 J (D) 120 J

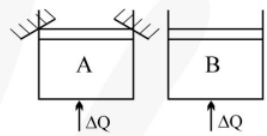
Q3. When a system is taken from state 'a' to state 'b' along the path 'acb', it is found that a quantity of heat Q = 200 J is absorbed by the system and a work W = 80J is done by it. Along the path 'adb', Q = 144J. The work done along the path 'adb' is



(A) 6J (B) 12 J

(C) 18 J (D) 24 J

Q4. Two identical vessels A & B contain equal amount of ideal monoatomic gas. The piston of A is fixed but that of B is free. Same amount of heat is absorbed by A & B. If B's internal energy increases by 100 J, then change in internal energy of A is :-



(A) 100 J (B) $\frac{500}{3}$ J

(C) 250 J (D) none of these

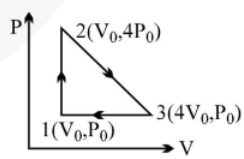
Q5. One mole of an ideal diatomic gas is taken through the cycle as shown in the figure.

$1 \rightarrow 2$: isochoric process

$2 \rightarrow 3$: straight line on P-V diagram

$3 \rightarrow 1$: isobaric process

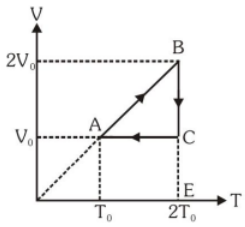
The average molecular speed of the gas in the states 1, 2 and 3 are in the ratio



(A) 1 : 2 : 2 (B) $1 : \sqrt{2} : \sqrt{2}$

(C) 1 : 1 : 1 (D) 1 : 2 : 4

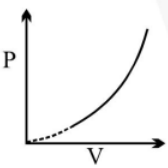
Q6. An ideal one mole monoatomic gas undergoes a cyclic process ABCA as shown in the fig. The ratio of heat absorbed during AB to the work done on the gas during BC is:–

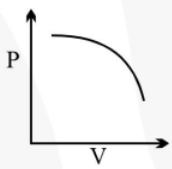


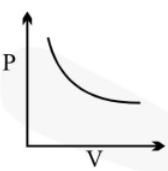
(A) $\frac{5}{2\ell n 2}$ (B) $\frac{5}{3}$

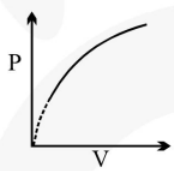
(C) $\frac{5}{4\ell n 2}$ (D) $\frac{5}{6}$

Q7. An ideal gas follows a process PT = constant. The correct graph between pressure & volume is -

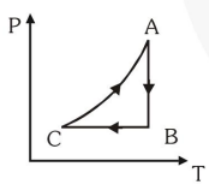
(A) 

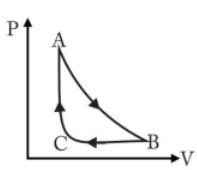
(B) 

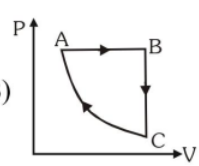
(C) 

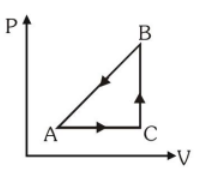
(D) 

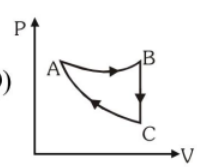
Q8. The P–T diagram for an ideal gas is shown in the figure, where AC is an adiabatic process, find the corresponding P–V diagram :–



(A) 

(B) 

(C) 

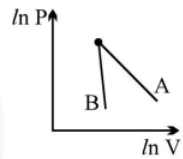
(D) 

Q9. During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its absolute temperature. The ratio C_P/C_V for the gas is-

(A) 4/3 (B) 2

(C) 5/3 (D) 3/2

Q10. The figure, shows the graph of logarithmic reading of pressure and volume for two ideal gases A and B undergoing adiabatic process. From figure it can be concluded that



(A) gas B is diatomic

(B) gas A and B both are diatomic

(C) gas A is monoatomic

(D) gas B is monoatomic & gas A is diatomic

Q11. An ideal gas at 27^0C is compressed adiabatically to 8/27 of its original volume. If $\gamma = 5/3$, then the rise in temperature is:-

(A) 450 K (B) 375 K

(C) 675 K (D) 405 K

Q12. Monoatomic, diatomic and triatomic gases whose initial volume and pressure are same, are compressed till their volume becomes half the initial volume.

(A) If the compression is adiabatic then monoatomic gas will have maximum final pressure.

(B) If the compression is adiabatic then triatomic gas will have maximum final pressure.

(C) If the compression is adiabatic then their final pressure will be same.

(D) If the compression is isothermal then their final pressure will be different.

Q13. A inert gas obeys the law PV^x = constant. For what value of x, it has negative molar specific heat–

(A) x > 1.67 (B) x < 1.67

(C) 1 < x < 1.4 (D) 1 < x < 1.67

Q14. Four moles of hydrogen, two moles of helium and one mole of water vapour form an ideal gas mixture. What is the molar specific heat at constant pressure of mixture?

(A) $\frac{16}{7}R$ (B) $\frac{23}{7}R$

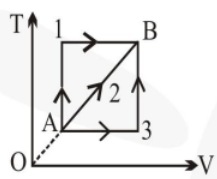
(C) $\frac{19}{7}R$ (D) $\frac{26}{7}R$

Q15. If γ be the ratio of specific heats of a perfect gas, the number of degrees of freedom of a molecule of the gas is:

(A) $\frac{25}{2}(\gamma-1)$ (B) $\frac{3\gamma-1}{2\gamma-1}$

(C) $\frac{2}{\gamma-1}$ (D) $\frac{9}{2}(\gamma-1)$

Q16. A given mass of a gas expands from a state A to the state B by three paths 1, 2 and 3 as shown in T-V indicator diagram. If W_1, W_2 and W_3 respectively be the work done by the gas along the three paths, then-



(A) $W_1 > W_2 > W_3$

(B) $W_1 < W_2 < W_3$

(C) $W_1 = W_2 = W_3$

(D) $W_1 < W_2$, $W_1 > W_3$

Q17. An ideal gas expands from volume V_1 to V_2. This may be achieved by either of the three processes: isobaric, isothermal and adiabatic. Let ΔU be the change in internal energy of the gas, Q be the quantity of heat added to the system and W be the work done by the gas . Identify which of the following statements is not true for ΔU?

(A) ΔU is least under adiabatic process.

(B) ΔU is greatest under adiabatic process.

(C) ΔU is greatest under the isobaric process.

(D) ΔU in isothermal process lies in-between the values obtained under isobaric and adiabatic processes.

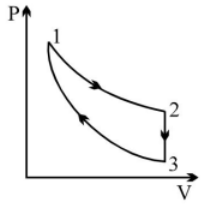
Q18. Three processes form a thermodynamic cycle as shown on P-V diagram for an ideal gas. Process $1 \rightarrow 2$ takes place at constant temperature (300K). Process $2 \rightarrow 3$ takes place at constant volume. During this process 40J of heat leaves the system. Process $3 \rightarrow 1$ is adiabatic and temperature T_3 is 275K. Work done by the gas during the process $3 \rightarrow 1$ is :-

(A) – 40J

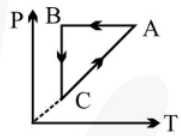
(B) – 20J

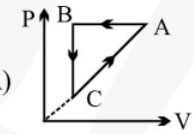
(C) +40J

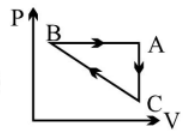
(D) +20J

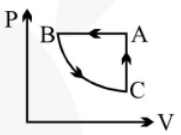


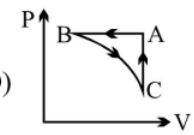
Q19. A cyclic process ABCA is shown in PT diagram. When presented on PV, it would



(A)


(B)


(C)


(D)


Q20. A carnot engine takes 3×10^6 cal of heat from a reservoir at 627 °C and gives it to a sink at 27 °C. The work done by the engine is-

(A) 4.2×10^6 J (B) 8.4×10^6 J

(C) 16.8×10^6 J (D) zero

Q21. A carnot engine, whose efficiency is 40%, takes heat from a source maintained at temperature of 500K. It is desired to have an engine of efficiency 60% then, the intake temperature for the same must be :

(A) 250 K (B) 750 K

(C) 350 K (D) 700 K

Q22. A reversible refrigerator operates between a low temperature reservoir at T_C and a high temperature reservoir at T_H. Its coefficient of performance is given by :

(A) $(T_H - T_C)/T_C$ (B) $T_C/(T_H - T_C)$

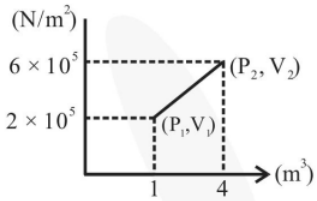
(C) $(T_H - T_C)/T_H$ (D) $T_H/(T_H - T_C)$

(E) $T_H(T_H + T_C)$

ONE OR MORE THAN ONE CORRECT TYPE

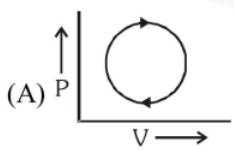
Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

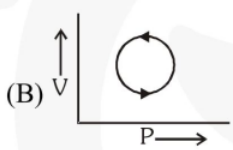
Q1. A system changes from the state (P_1, V_1) to (P_2, V_2) as shown in the figure below. What is the work done by the system :-

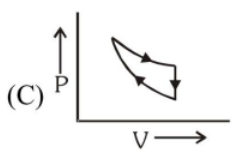


(A) 7.5×10^5 Joule (B) 7.5×10^4 erg
(C) 12×10^5 Joule (D) 6×10^5 Joule

Q2. The following are the P–V diagrams for cyclic processes for a gas. In which of these processes is heat absorbed by the gas ?

(A) 

(B) 

(C) 

(D) 

Q3. During the melting of a slab of ice at 273 K at atmospheric pressure:–

(A) Positive work is done by the ice–water system on the atmosphere.

(B) Positive work is done on the ice–water system by the atmosphere.

(C) The internal energy of ice–water system increases.

(D) The internal energy of the ice–water system decreases.

Q4. A vertical cylinder with heat-conducting walls is closed at the bottom and is fitted with a smooth light piston. It contains one mole of an ideal gas. The temperature of the gas is always equal to the surrounding's temperature T_0. The piston is moved up slowly to increase the volume of the gas to η times. Which of the following is incorrect?

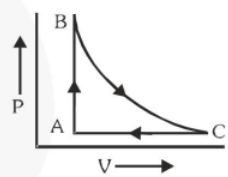
(A) Work done by the gas is $RT_0 \ln \eta$.

(B) Work done against the atmosphere is $RT_0(\eta - 1)$.

(C) There is no change in the internal energy of the gas.

(D) The final pressure of the gas is $\frac{1}{(\eta - 1)}$ times its initial pressure.

Q5. One mole of monoatomic ideal gas undergoes a cyclic process ABCA as shown in figure. Process BC is adiabatic. The temperatures at A, B and C are 300, 600 and 450K respectively. Choose the correct statement(s).



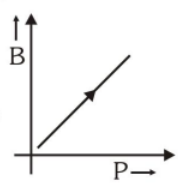
(A) In process CA change in internal energy is 225R.

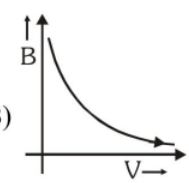
(B) In process AB change in internal energy is –150R.

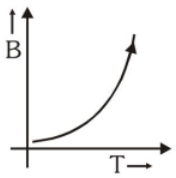
(C) In process BC change in internal energy is –225R.

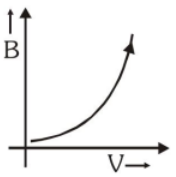
(D) Change in internal energy during the whole cyclic process is +150R.

Q6. A sample of gas follows process represented by PV^2 = constant. Bulk modulus for this process is B, then which of the following graph is correct?

(A) 

(B) 

(C) 

(D) 

Q7. An ideal gas expands in such a way that PV^2 = constant throughout the process.

(A) The graph of the process of T-V diagram is a parabola.

(B) The graph of the process of T-V diagram is a straight line.

(C) Such an expansion is possible only with heating.

(D) Such an expansion is possible only with cooling.

Q8. A gas expands such that its initial and final temperatures are equal. Also, the process followed by the gas traces a straight line on the P–V diagram :-

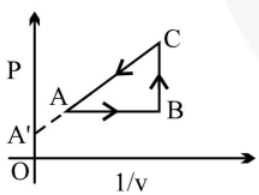
(A) The temperature of the gas remains constant throughout.

(B) The temperature of the gas first increases and then decreases.

(C) The temperature of the gas first decreases and then increases.

(D) The straight line has a negative slope.

Q9. An enclosed ideal gas is taken through a cycle as shown in the figure. Then



(A) Along AB, temperature decreases while along BC temperature increases

(B) Along AB, temperature increases while along BC the temperature decreases.

(C) Along CA work is done by the gas and the internal energy remains constant.

(D) Along CA work is done on the gas and internal energy of the gas increases.

Q10. An ideal gas can be expanded from an initial state to a certain volume through two different processes :

(i) PV^2 = constant and

(ii) $P = KV^2$ where K is a positive constant.

Then:–

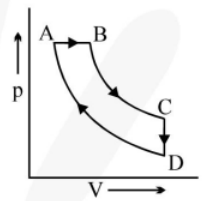
(A) Final temperature in (i) will be greater than in (ii)

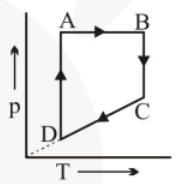
(B) Final temperature in (ii) will be greater than in (i)

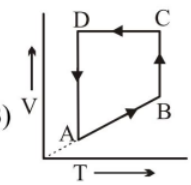
(C) Total heat given to the gas in (i) case is greater than in (ii)

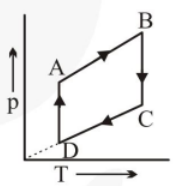
(D) Total heat given to the gas in (ii) case is greater than in (i)

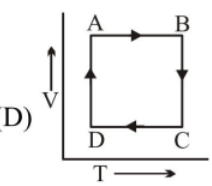
Q11. A cyclic process ABCD is shown in the p-V diagram. Which of the following curves represents the same process if BC & DA are isothermal processes



(A) 

(B) 

(C) 

(D) 

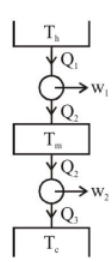
Q12. Suppose that two heat engines are connected in series, such that the heat exhaust of the first engine is used as the heat input of the second engine as shown in figure. The efficiencies of the engines are η_1 and η_2, respectively. The net efficiency of the combination is given by :

(A) $\eta_{net} = \eta_2 + (1-\eta_1)\eta_2$

(B) $\eta_{net} = \dfrac{\eta_1}{(1-\eta_1)\eta_2}$

(C) $\eta_{net} = \eta_1 + (1-\eta_1)\eta_2$

(D) $\eta_{net} = \dfrac{1-\eta_1}{(1-\eta_2)\eta_2}$

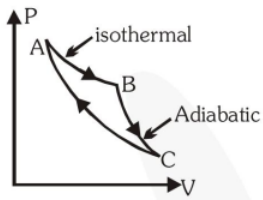


PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 13 & 15

A cyclic process for an ideal gas is shown in figure. Given $W_{AB} = +700$ J, $W_{BC} = +400$ J, $Q_{CA} = -100$J.



Q13. Find ΔU_{BC}

(A) –700 J (B) –400 J

(C) –100 J (D) 400 J

Q14. Find W_{CA}

(A) –500 J (B) 500 J

(C) 400 J (D) –400 J

Q15. The efficiency of the cycle is -

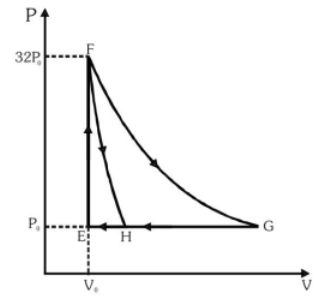
(A) 100% (B) 83.44%

(C) 85.71% (D) 81.11%

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching For each question, choose the option corresponding to the correct matching.*

Q16. One mole of a monatomic ideal gas is taken along two cyclic processes E→F→G→E and E→F→H→E as shown in the PV diagram. The processes involved are purely isochoric, isobaric, isothermal or adiabatic.



Match the paths in List I with the magnitudes of the work done in List II and select the correct answer using the codes given below the lists.

List I	**List II**
P. G → E	1. 160 $P_0 V_0$ ln2
Q. G → H	2. 36 $P_0 V_0$
R. F → H	3. 24 $P_0 V_0$
S. F → G	4. 31 $P_0 V_0$

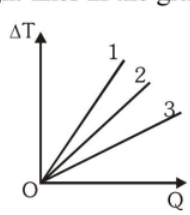
Codes :

	P	**Q**	**R**	**S**
(A)	4	3	2	1
(B)	4	3	1	2
(C)	3	1	2	4
(D)	1	3	2	4

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q17. The straight lines in the figure depict the variations in temperature ΔT as a function of the amount of heat supplied Q in different process involving the change of state of a monoatomic and a diatomic ideal gas. The initial states (P,V,T) of the two gases are the same. Match the processes as described, with the straight lines in the graph as numbered.



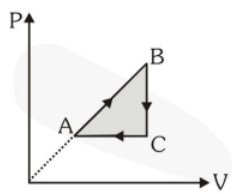
Column-I	Column-II
(A) Iosbaric process of monoatomic gas.	(p) 1
(B) Isobaric process of diatomic gas	(q) 2
(C) Isochoric process of monoatomic gas	(r) 3
(D) Isochoric process of diatomic gas	(s) x-axis (i.e. 'Q' axis)

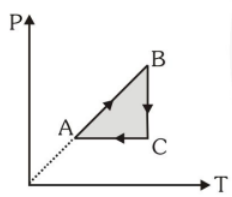
Q18. For one mole of a monoatomic gas :–

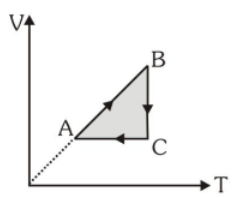
Column I	Column II
(A) Isothermal bulk modulus	(p) $-\frac{RT}{V^2}$
(B) Adiabatic bulk modulus	(q) $-\frac{5P}{3V}$
(C) Slope of P–V graph in isothermal process	(r) T/V
(D) Slope of P–V graph in adiabatic process	(s) 4T/3V
	(t) None

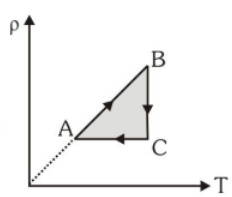
Q19. For a ideal monoatomic gas match the following graphs for constant mass in different processes (ρ = Density of gas)

Column I

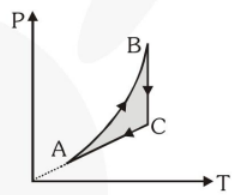
(A)


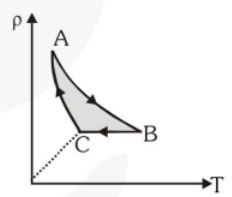
(B)


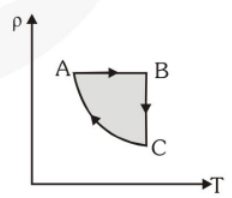
(C)


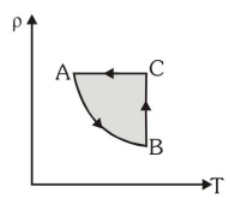
(D)


Column II

(p)


(q)


(r)


(s)


NUMERICAL TYPE

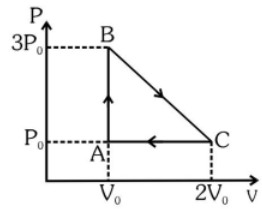
*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q20. A carnot engine working between 400 K and 800 K has a work output of 1200 J per cycle. What is the amount of heat energy supplied to the engine from source per cycle?

Q21. The temperature insides & outside of refrigerator are 260 K and 315 K respectively. Assuming that the refrigerator cycle is reversible, calculate the heat delivered to surroundings for every joule of work done.

SUBJECTIVE TYPE

Q22. One mole of an ideal monoatomic gas is taken round the cyclic process ABCA as shown in fig. calculate.



(a) The work done by the gas.

(b) The heat rejected by the gas in the path CA and heat absorbed in the path AB.

(c) The net heat absorbed by the gas in the path BC.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
A	D	D	A	D	B	C	B	C	C
11	12	13	14	15	16	17	18	19	
D	B	B	C	A	C	A	C	A	

Ex-1A

1. A **2.** A,C **3.** B **4.** B **5.** B,C
6. C **7.** C,D **8.** A,B,C,D **9.** B **10.** A
11. C **12.** C **13.** A **14.** B
15. (A)→ q; (B)→ r; (C)→ p; (D)→ s
16. (A)→ p; (B)→ q; (C)→ q
17. A→ B 6000 J, B→ C is zero, C→ D –1000 J, and D→ A is Zero
18. $\frac{3}{2}R$ (monoatomic) & degree of freedom f = 3

(b) $C_v = \frac{3}{2}R$ and $\gamma = \frac{5}{3}$, (c) 450R
19. (a) 24×10^2 J, (b) 36×10^2 J, (c) 60×10^2 J
20. 3.36×10^6 J

Ex-2									
1	2	3	4	5	6	7	8	9	10
C	A	D	B	A	C	C	A	D	D
11	12	13	14	15	16	17	18	19	20
B	A	D	B	C	A	B	A	C	B
21	22								
B	B								

Ex-2A

1. C **2.** A,B,C **3.** B,C **4.** D **5.** C
6. A,B,C **7.** D **8.** B,D **9.** A **10.** A,C
11. A,B **12.** C **13.** B **14.** A **15.** C
16. A
17. (A)→ q; (B)→ r; (C)→ p; (D)→ q
18. (A)→ t; (B)→ t; (C)→ p; (D)→ q
19. (A)→ s; (B)→ r; (C)→ q; (D)→ p
20. 2400 joule per cycle
21. 5.7 J
22. (a) P_0V_0, (b) $\frac{5}{2}P_0V_0$, $3P_0V_0$, (c) $\frac{P_0V_0}{2}$

-: Important Notes & Formulas :-